Assignment 1: Book Management System (C++)

Objective

To implement **Class and Object concepts** using:

- Constructor
- Accept method
- Display method
- Getter and Setter methods
- Simple logical operations

Problem Statement

Create a C++ program to manage **Book Details** in a library using Object-Oriented Programming principles.

🔷 **Requirements:**

1️⃣ **Create a class named Book.**

2️⃣ **Private Data Members:**

- Book ID (int)
- Book Title (string)
- Author Name (string)
- Price (float)
- Quantity (int)

🔷 **Member Functions:**

✅ **1. Constructor**

- Default constructor should initialize:
  - Book ID = 0
  - Book Title = "Not Available"
  - Author Name = "Unknown"

    - Price = 0.0
    - Quantity = 0
- (Optional ⭐) Parameterized constructor.

✅ **2. Accept Function**

- accept() → Accept book details from user.

✅ **3. Display Function**

- display() → Display complete book details.

✅ **4. Getter Methods**

- getBookId()
- getTitle()
- getAuthor()
- getPrice()
- getQuantity()

✅ **5. Setter Methods**

- setBookId(int id)
- setTitle(string title)
- setAuthor(string author)
- setPrice(float price)
- setQuantity(int qty)

✅ **6. Additional Function**

- calculateTotalValue()
    - Total Value = Price × Quantity

Main Function Requirements:

1. Create one object of class Book.
2. Call accept() and display().
3. Modify price using setter.
4. Print updated price using getter.
5. Display total stock value using calculateTotalValue().

Sample Output Format:

Enter Book ID: 201

Enter Book Title: C++ Programming

Enter Author Name: Bjarne Stroustrup

Enter Price: 500

Enter Quantity: 10

Book Details:

ID: 201

Title: C++ Programming

Author: Bjarne Stroustrup

Price: 500

Quantity: 10

Total Stock Value: 5000

Here's one more assignment with a slightly different domain 👇

Assignment 2: Vehicle Information System (C++)

Objective

To implement **Class and Object concepts** using:

- Constructor
- Accept method
- Display method
- Getter and Setter methods
- Simple calculation logic

Problem Statement

Create a C++ program to manage **Vehicle Details** using Object-Oriented Programming principles.

🔷 **Requirements:**

1️⃣ **Create a class named Vehicle.**

2️⃣ **Private Data Members:**

- Vehicle Number (string)
- Brand (string)
- Model (string)
- Price (float)
- Mileage (float)

🔷 **Member Functions:**

✅ **1. Constructor**

- Default constructor should initialize:
    - Vehicle Number = "Not Registered"

    - Brand = "Unknown"
    - Model = "Unknown"
    - Price = 0.0
    - Mileage = 0.0
- (Optional ⭐) Parameterized constructor.

✅ 2. Accept Function

- accept() → Accept vehicle details from user.

✅ 3. Display Function

- display() → Display vehicle details.

✅ 4. Getter Methods

- getVehicleNumber()
- getBrand()
- getModel()
- getPrice()
- getMileage()

✅ 5. Setter Methods

- setVehicleNumber(string num)
- setBrand(string brand)
- setModel(string model)
- setPrice(float price)
- setMileage(float mileage)

✅ 6. Additional Function

- calculateResaleValue()
  - Resale Value = 80% of Price

Main Function Requirements:

1. Create one object of class Vehicle.
2. Call accept() and display().
3. Update price using setter.
4. Print updated price using getter.
5. Display resale value.

Sample Output Format:

Enter Vehicle Number: MH12AB1234

Enter Brand: Honda

Enter Model: City

Enter Price: 1000000

Enter Mileage: 18.5

Vehicle Details:

Number: MH12AB1234

Brand: Honda

Model: City

Price: 1000000

Mileage: 18.5

Resale Value: 800000

Assignment 6: Product Inventory System (With Static Members) – C++

Objective

To implement:

- Class and Object
- Constructor
- Accept & Display methods
- Getter and Setter methods
- **Static Data Member**
- **Static Member Function**

Problem Statement

Create a C++ program to manage **Product Details** in a store using Object-Oriented Programming principles.

🔷 **Requirements:**

1️⃣ **Create a class named Product.**

🔷 **Private Data Members:**

- Product ID (int)
- Product Name (string)
- Price (float)
- Quantity (int)

🔷 **Static Data Member:**

- static int productCount;
    - This variable should count how many Product objects are created.

🔷 **Member Functions:**

✅ **1. Constructor**

- Default constructor should:
    - Initialize data members.
    - Increment productCount every time an object is created.
- (Optional ⭐) Parameterized constructor.

✅ **2. Accept Function**

- accept() → Accept product details from user.

✅ **3. Display Function**

- display() → Display product details.

✅ **4. Getter Methods**

- getProductId()
- getProductName()
- getPrice()
- getQuantity()

✅ **5. Setter Methods**

- setProductId(int id)
- setProductName(string name)
- setPrice(float price)
- setQuantity(int qty)

✅ **6. Static Member Function**

- static void showProductCount()
    - This function should display the total number of Product objects created.
    - It should access only static members.

✅ **7. Additional Function**

- calculateTotalValue()
    - Total Value = Price × Quantity

🔷 **Important Instruction:**

- Define and initialize the static data member **outside the class**.

Example:

int Product::productCount = 0;

Main Function Requirements:

1. Create at least **3 Product objects**.
2. Call accept() and display() for each.
3. Call Product::showProductCount() using class name.
4. Display total stock value of each product.

Sample Output Format:

Enter details for Product 1

Enter Product ID: 101

Enter Product Name: Laptop

Enter Price: 50000

Enter Quantity: 2

Enter details for Product 2

Enter Product ID: 102

Enter Product Name: Mouse

Enter Price: 500

Enter Quantity: 10

Total Products Created: 2